Lesson 8.2: Non-zero-sum games

Introduction to game theory

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Non-strictly-competitive games

Example (Coconut game)

Alice (A) and Bob (B) are at the base of a tree which has only one coconut. The coconut provides 10 kilocalories (kc) in total energy. To get the coconut, one (or both) must climb and drop it to crack it open. Because Alice is heavy, it costs her 2kc in effort to climb, whereas Bob is light and his effort is negligible.

If Bob climbs (c) the tree and Alice waits (W) below then Alice will get to the coconut first, eating most of it (9kc) and leaving only a small portion for Bob. If Alice climbs (C) and Bob waits (w) below then Bob will get to it first and eat his fill (4kc worth) before Alice climbs down and eats the rest. If both climb up, Bob will climb down quicker and eat some (3kc worth) before Alice climbs down.

Simultaneous moves; cooperation

Alice:

	c	w
C	5	4
W	9	0

Bob:

	c	w
C	3	4
W	1	0

- how should Alice and Bob play?
- combined:

	c	w
C	5, 3	4, 4
W	9, 1	0, 0

Questions

- what is a 'fair' outcome?
- for what outcome/payoffs should players 'settle'?

Alice moves first

- Bob has information about Alice's move; Alice knows this

	c	c	w	w
	c	w	c	w
C	5, 3	5, 3	4, 4	4, 4
W	9, 1	0, 0	9, 1	0, 0

reduces to:

	w
	c
C	4, 4
W	9, 1

. . . and to:

	w
	c
W	9, 1

Solution

$\begin{pmatrix} & w \\ W, & c \end{pmatrix}$; values: 9, 1.